INCH-POUND

MIL-DTL-432H
21 JULY 2008
SUPERSEDING
MIL-G-432G
21 November 1989

DETAIL SPECIFICATION

GASKETS, NONMETALLIC, SYNTHETIC RUBBER

INACTIVE FOR NEW DESIGN AFTER 31 AUGUST 1998

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers five types of synthetic rubber, nonmetallic gaskets.

1.2 Classification. The gaskets are of the following types and classes as specified (see 6.2):

Round cross section (see figure 1).

Type I - 2 inch bung.

Type I - .75 inch vent.

Rectangular cross section on 55-gallon drums or steel drums. (see figure 2).

Type II - 2 inch bung.

Type II - .75 inch vent.

Beaded edge 2.25 inch ID by 3.375 inch OD for 5-gallon gasoline can with carrying handle (see figure 3).

Type III Class 1 - For use at -40°F (-40°C) and above.

Type III Class 2 - For use at -60°F (-51°C) and above.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Philadelphia (DSCP), ATTN: DSCP-NASA, 700 Robbins Avenue., Philadelphia, PA 19111-5096 or e-mail to dscpg&ispecomments@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 5330

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Rectangular cross section for steel drums (see figure 4).

Type VI - 2 inch bung.

Type VI - .75 inch vent.

Rectangular cross section for filling tube, flexible nozzle cam type (see figure 5).

Type VII - 1.5 inch ID by 2.5 inch 0D

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in section 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY NATICK SOLDIER CENTER

8-1-161 - Pressure Jig Assembly for Gasket, Nonmetallic

(Copies of these drawings are available from the U.S. Army Natick Soldier Center, ATTN: AMSRD-NSC-AD-B, Natick, MA 01760.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 395 Rubber Property - Compression Set
D 471 Rubber Property - Effect of Liquids
D 1053 Rubber Property - Stiffening at Low Temperatures: Flexible Polymers and Coated Fabrics
D 1229 Rubber Property - Compression Set at Low Temperatures
D 2240 Rubber Property - Durometer Hardness

(Copies of this document are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959.)

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AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4 Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from www.asq.org or the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material. Gaskets shall be made of various types of synthetic rubber that will meet the requirements of this specification. Phosphonitrilic fluoroelastomers have been found suitable for the type III, class 2 gaskets.

3.3 Design and dimensions. Design and dimensions of each type and class of gasket shall be in accordance with figures 1 to 5, as applicable, and shall be furnished in the size required (see 6.2).

3.4 Physical characteristics.

3.4.1 Hardness (Durometer). The gasket material (slabs of synthetic rubber from which the gaskets are made) shall have a Durometer hardness as shown in table I when tested as specified in 4.3.1.1. -

TABLE I. Hardness

Type	Hardness (Durometer)
I, VI	60-70
II	75-85
III	65-75
VII	40-50

3.5 Performance.

3.5.1 Volume change. The slabs of synthetic rubber material from which the gaskets are made shall not swell more than 40 percent, and there shall be no shrinkage when tested as specified in 4.3.1.1.

3.5.2 Weight change. Slabs of synthetic rubber material from which types I, II, III, and VI gaskets are made shall not change more than 25 percent from the original weight when tested as specified in 4.3.1.1.

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3.5.3 Immersion resistance. The gaskets shall not disintegrate, crack, nor stretch or distort more than 5 percent or show shrinkage of more than 2 percent of the original inside diameter when tested as specified in 4.3.4. In addition, after 24 hours drying time, the gasket shall not show signs of disintegration, stretching, shrinking, cracking, or distortion when compared with new untested gaskets of the same lot sample.

3.6 Low temperature compression set. The low temperature compression set shall not exceed 60 percent when tested as specified in 4.3.4.

3.6.1 Torsional modulus ratio (type III only). The torsional modulus ratio shall not exceed 10 when tested as specified in 4.3.1.1.

3.7 Compression set. The compression set shall not exceed 40 percent of original deflection when tested as specified in 4.3.4.

3.8 Endurance (except for type VII). Gaskets shall withstand the test specified in 4.3.4.

3.9 Leakage (except for type VII). Gaskets shall withstand the test specified in 4.3.4.

3.10 Strength. Gaskets either wet or dry shall not fracture, break, nor distort beyond the periphery of the applicable gasket seat closure plug, can closure plug, or can closure when tested as specified in 4.3.4.

3.11 Replacement of defective components. During the manufacturing process, components having material defects or damages that are classified as defects in 4.3.2 and 4.3.3 shall be removed from production and replaced with nondefective and properly matched components.

3.12 Workmanship. The end items shall conform to the quality of product established by this specification. The finished gaskets shall be clean and free from defects, such as rough or scratched surfaces and edges, cuts, holes, open blisters, and lumps. Flash resulting from mold vulcanization will be acceptable within the limits shown on figure 1.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection- (see 4.3).

4.2 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.3.2 and 4.3.3 and tested for the characteristics specified in 4.3.4.

4.3 Conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with ASQ Z1.4.

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4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicab1e purchase document. In addition, testing shall be performed on components and materials listed in table II.

4.3.1.1 Component testing. All test reports shall contain the individual values utilized in expressing the final result. All requirements are applicable to the sample unit. The sample unit shall be one slab of the synthetic rubber, 6 inches by 6 inches (15.24cm by 15.24cm). Thickness of the sample unit shall be determined by the type gasket specified, be uniform in thickness to .005 inches (.127mm) and shall be identical in composition and cure to the inspection lot represented. The samples selected shall be suitably marked and dated for identification with the inspection lot. Samples for testing may be selected from the stock material from-which the end item is made at the time of manufacture. The stock material shall be unacceptable if one or more sample units fail to meet any test requirement specified. The sample size shall be as follows:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. Component testing

Material	Characteristic	Requirement paragraph	Test method
Synthetic rubber	Hardness (Durometer)	3.4.1	4.4.1
	Volume change	3.5.1	4.4.2
	Weight change	3.5.2	4.4.2 and 4.4.3
	Torsional modulus ratio (type III only):		
	Original	3.6.1	4.4.4.1
	After solvent extraction	3.6.1	4.4.4.2

4.3.2 End item visual examination. The end-items shall be examined for the defects listed in table III. The lot shall be expressed in units of gaskets of the same type. The sample unit shall be one gasket of one type. The inspection level shall be I in accordance with ASQ Z1.4 (see 6.4).

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TABLE III. End item visual defects

Examine	Detect
Design	Not in accordance with specified requirements
Appearance and workmanship	Not clean Rough edges or rough surface Edges or surface nicked, scratched, or abraded Any cut or hole Any imbedded foreign matter Any open blister Any lump or closed blister

4.3.3 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified in figures 1 through 5, as applicable. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of gaskets, of the same type. The sample unit shall be one gasket, of one type. The inspection level shall be S-3 in accordance with ASQ Z1.4 (see 6.4).

4.3.4 End item testing. The end items shall be tested for the characteristics listed in table IV. The lot size shall be expressed in units of gaskets, of the same type. The sample unit- shall be 5 gaskets, of the same type. All test requirements shall be applicable to the sample unit. All test reports shall contain the individual values used to express the final result. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The sample size shall be as follows:

Lot size (gaskets)	Sample Size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE IV. End item tests

Characteristics	Requirement paragraph	Test method	Number determinations per sample unit	Results reported as pass or fail
Immersion resistance	3.5.3	4.4.2.1	1	X
Immersion resistance(after 24 hours)	3.5.3	4.4.2.1	1	X
Low temperature compression set. (All gaskets except type III)	3.6	4.4.4	1	X
Low temperature compression set. (For type III gasket only):				
Original	3.6	4.4.4.3	1	X
After solvent extraction	3.6	4.4.4.4	1	X
Compression set	3.7	4.4.5	1	X
Endurance	3.8	4.4.6	1	X
Leakage	3.9	4.4.7	1	X
Strength	3.10	4.4.6	1	X

4.4 Methods of inspection.

4.4.1 Hardness test. The hardness shall be determined in accordance with ASTM D 2240.

4.4.2 Immersion medium test. The immersion medium for volume change and weight change shall be fuel C in accordance with ASTM D 471.

4.4.2.1 Immersion resistance test. The gaskets shall be immersed in the immersion medium specified in 4.5.2 for 46 ± 1/4 hour at 73.4° ± 3.6°F.(23° ± 1.1°C). Measurements shall be made using an inside divider and a micrometer caliper calibrated in thousands. After 24 hours of drying time at 73.4° ± 3.6°F (23° ± 1.1°C), examination shall be made for disintegration, stretching, shrinking, cracking, or distortion.

4.4.3 Weight change test. The weight change shall be determined by using the weights of specimen in air recorded in 4.4.2 and using the method of calculation specified in ASTM D 471 to determine the change in weight.

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4.4.4 Low temperature compression set test. Low temperature compression set shall be determined in accordance with ASTM D 1229. The specimens shall be compressed 40 percent and exposure time shall be 94 ± 1/2 hour at -40°F (-40°C). Readings shall be taken after the 30-minute recovery period.

4.4.4.1 Torsional modulus ratio (material for type III gaskets). The torsional modulus ratio shall be determined in accordance with ASTM D 1053. The material to be used for type III, class 1 gaskets shall be tested at -40° ± 2°F (-40° ± 2°C). The material to be used for type III, class 2 gaskets shall be tested at -60° ± 2°F (-51° ± 2°C). The liquid coolant shall be methyl alcohol, and exposure time shall be five minutes.

4.4.4.2 Torsional modulus ratio after solvent extraction (material for type III gaskets). The gasket material shall be immersed in Reference Fuel B, ASTM D 471, for 24 ± 1/4 hours at 73° ± 5°F (22.7° ± 1.6°C). The extracted sample shall then be removed from Fuel B, placed on a wire screen and dried with forced air for 24 ± 1/4 hours at 73° ± 5°F (22.7° ± 1.6°C). The torsional modulus ratio shall be determined as specified in 4.4.4.1.

4.4.4.3 Low temperature compression set test (type III gaskets). The low temperature compression set shall be determined in accordance with ASTM D 1229. The type III, class 1 gasket sample shall be compressed 40 percent and exposure time shall be 94 ± 1/2 hours at -40° ± 2°F (-40° ± 2°C). The type III, class 2 gasket sample shall be compressed 40 percent and the exposure time shall be 94 ± 1/2 hours at 60° ± 2°F (-51° ± 2°C). Readings shall be taken after the 30-minute recovery period.

4.4.4.4 Low temperature compression set after solvent extraction test (type III gaskets). The gasket shall be immersed in Reference Fuel B, ASTM D 471, for 24 ± 1/4 hours at 73° ± 5°F (22.7° ± 1.6°C). The sample shall then be removed from Fuel B, placed on a wire screen and dried with forced air for 24 ± 1/4 hours at 73°± 5°F (22.7°± 1.6°C). The low temperature compression set shall be determined according to 4.4.4.3.

4.4.5 Compression set test. The compression set test shall be performed in accordance with ASTM D 395, method B, except that the sample specimen thickness shall be the actual thickness of the gasket. Spacer bars shall be the thickness required to compress the gasket 25 ± 3 percent of the original gasket thickness. The test specimen shall be held compressed for 22 hours at 158° ± 2°F (70°± .9°C).

4.4.6 Endurance test (except type VII). A closure flange shall be thoroughly wetted with the special testing fuel specified in 4.5.2, a gasket placed on it, and a plug inserted. The plug shall be tightened to the torque shown in table V with a torque wrench and loosened 30 times. An interval of 6 to 7 seconds shall be allowed between each tightening and loosening. The plug shall then be tightened once more with a torque equal to 2-1/2 times that indicated in table V.

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TABLE V. Torque loading

Type	Size (inches)	Torque (foot-pounds)	Size (inches)	Torque (foot- pounds)
I	2	15	.75	7
II	2	35	.75	20
III	2	10	--	--
VI	2	15	.75	7

4.4.7 Leakage tests (except type VII). The leakage tests in the following paragraphs are to determine gasket failure by pressure loss. A pressure drop to 20 pounds per square inch (psi) (.138 MPa) or less reading on the gage under conditions as specified in the following test procedures shall be considered a failure to comply with the leakage requirement.

4.4.7.1 Inverted jig procedure (gasket in contact with fuel). The gasket subjected to the test specified in 4.4.7 shall be transferred to a pressure jig as shown on Drawing 8-1-161. One hundred milliliters of the special testing fuel specified in 4.5.2 shall be put in the container of the jig, and the plug with the gasket in place shall be inserted and tightened to the torque specified in table V. The jig shall be inverted so that the testing fuel is in contact with the plug. Air, or nitrogen under pressure, shall then be introduced through the air inlet until the pressure gage reads 25 psi (.172 MPa), the pressure being maintained by closing the needle valve on the air inlet. If a pressure drop to 20 psi (.138 MPa) or less on the gage indicator is observed within 10 minutes, the test shall be considered a failure. If the gasket passes the test, proceed with the test in 4.4.7.2.

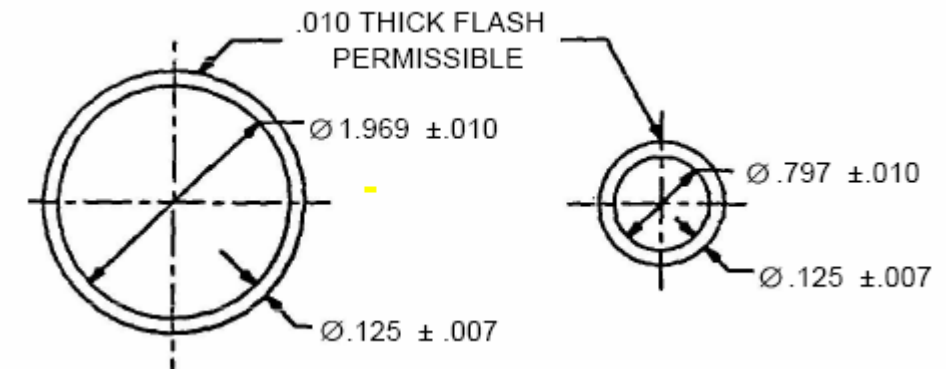
4.4.7.2 Upright jig procedure (gasket not in contact with fuel). The test shall be repeated with the same gasket in place (i.e., loosen, tighten to required torque, and introduce air again as in 4.5.7.1 with the exception that the jig shall not be inverted). If a pressure drop to 20 psi (.138 MPa) or less on the gage indicator is observed within 10 minutes, the test shall be considered a failure.

4.4.7.2.1 Room temperature. If no leakage occurs in 4.4.7.2, the pressure jig with gasket in place shall be placed in an upright position in a constant air temperature of 77º± 2°F (25º±.6ºC) and accurate pressure shall be noted. The jig assembly shall remain at this temperature for 24 hours. If during the 24 hour period a pressure drop to 20 psi (.138 MPa) or less on the gage indicator is observed, the test shall be considered a failure.

4.4.7.2.2 Water immersion. Continuing with the same assembly as specified in 4.4.7.2.1, it shall be placed under water at a controlled temperature of 140° ± 2°F (60º±.9ºC) and left for 24 hours. The assembly shall then be removed and allowed to remain at 77º± 2°F (25º±.6ºC) for 3 hours. If a pressure drop to 20 psi (.138 MPa) or less on the gage indicator is observed, the test shall be considered a failure.

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4.4.7.2.3 Cold temperature. Continuing with the same assembly, the pressure shall be observed and then the assembly shall be placed in a cold chamber at -40° ± 2°F (-40 ± 2°C) for 2 hours. A check shall be made for leakage of the testing fuel. The pressure jig shall then be removed from the chamber and allowed to remain at 77°± 2°F (25°±.6°C) for 3 hours. If a pressure drop to 20 psi (.138 MPa) or less on the gage is observed, the test shall be considered a failure.

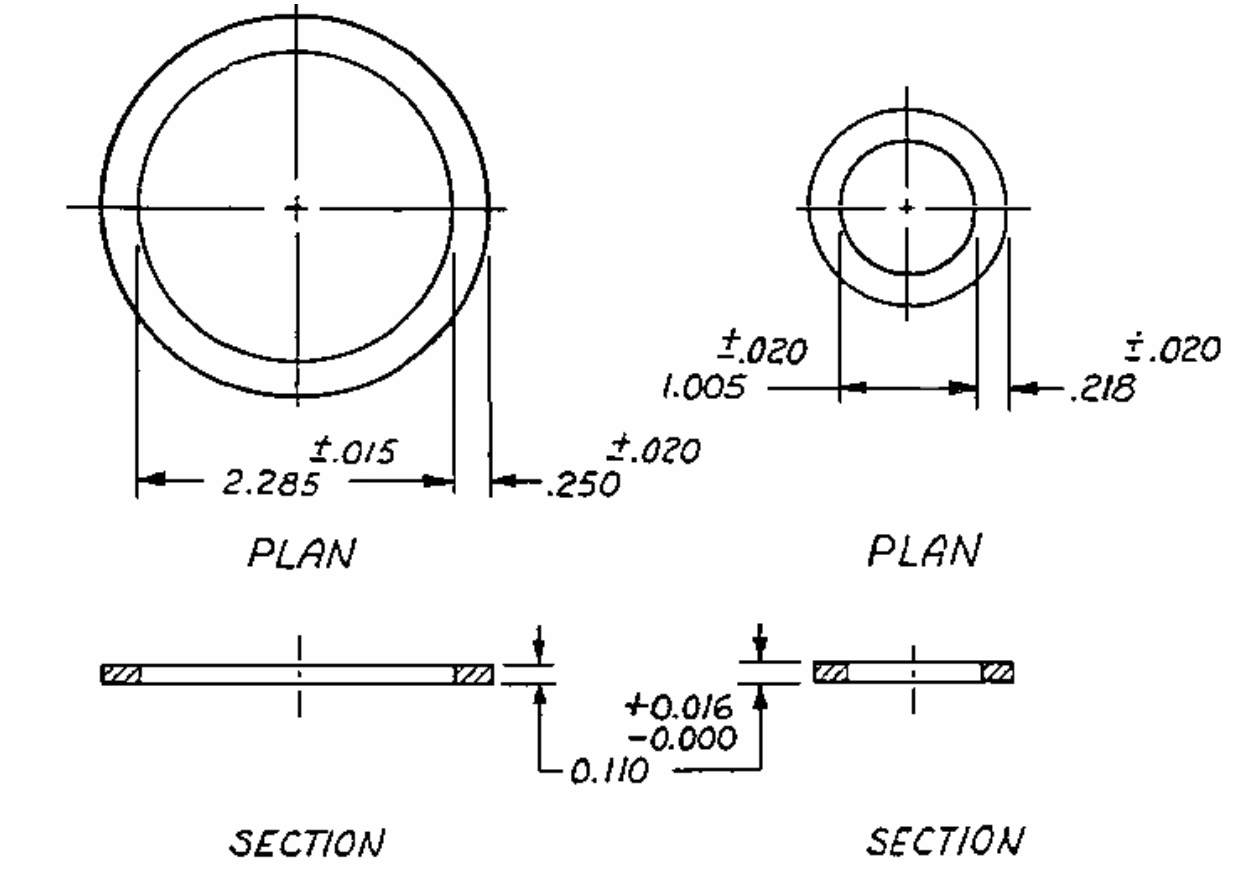


Type I – 2 inch Bung

Type I - .75 inch Vent

FIGURE 1. TYPE 1 GASKETS.

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Type II – 2 inch Bung **Type II - .75 inch Vent**

FIGURE 2. Type II Gaskets.

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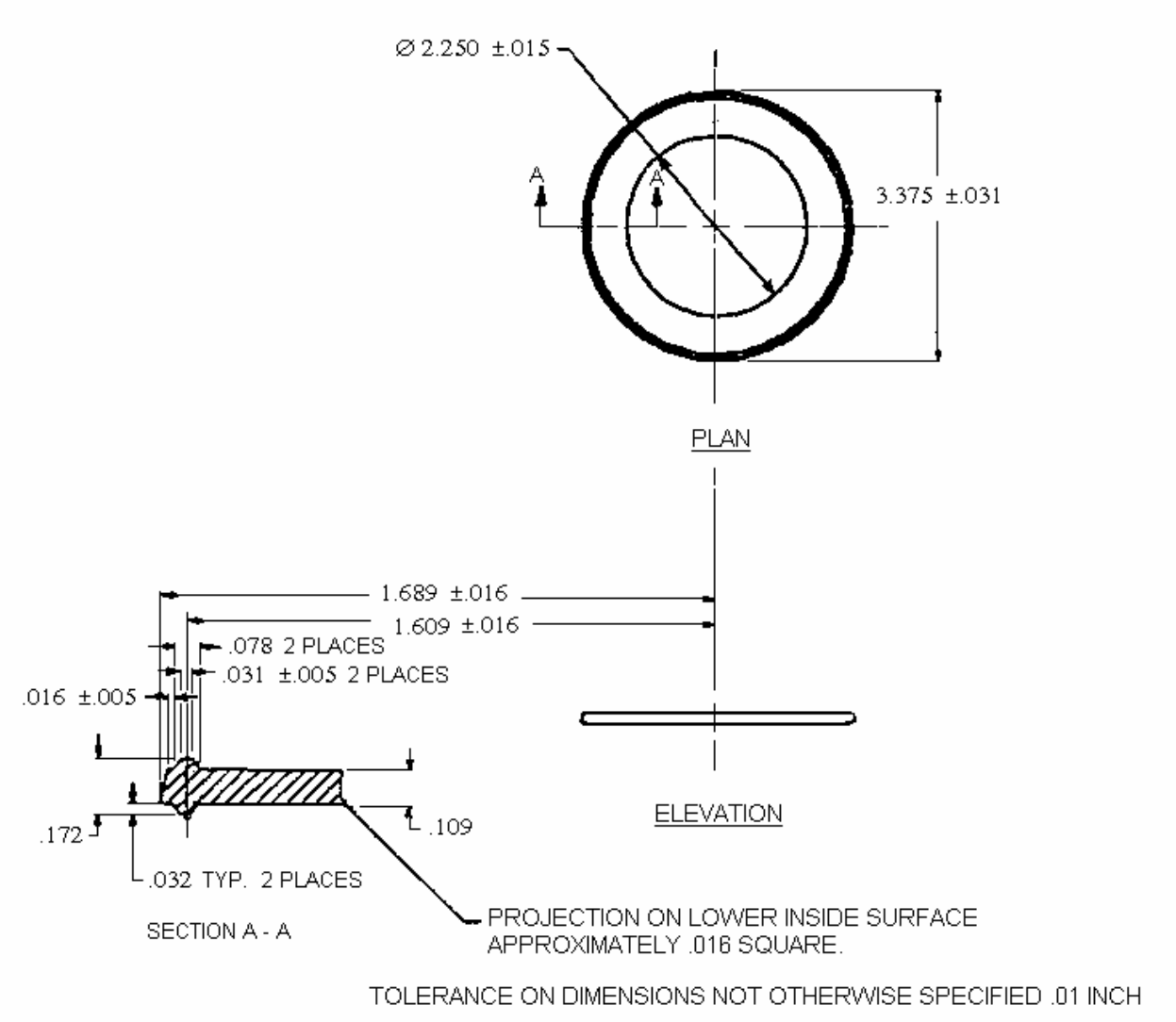
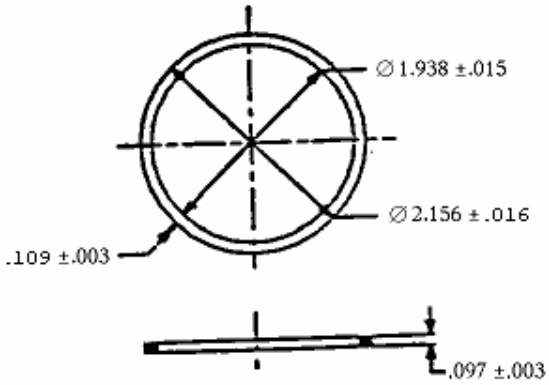


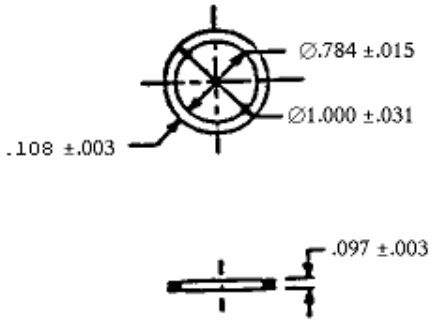
FIGURE 3. Type III Gaskets.

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NOTES:



Type VI - 2 inch Bung



Type VI - .75 inch Vent

FIGURE 4. Type VI Gaskets.

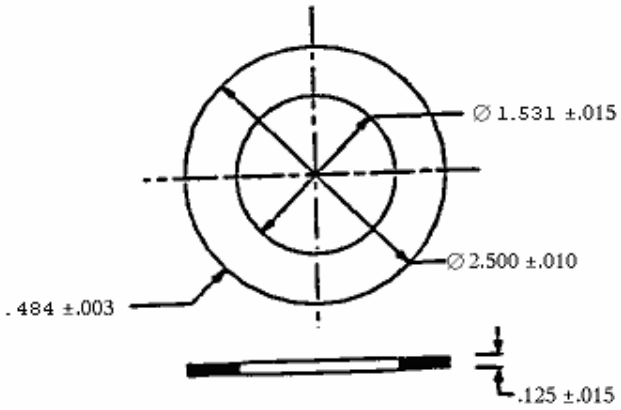


FIGURE 5. Type VII Gaskets.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Services or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

(This Section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The gaskets are intended for use with can type flexible nozzle filling tube, drums, and other containers used for transporting aromatic gasoline, or other petroleum fractions. The type III, class 1 gaskets are intended for use at -40°F (-40°C) and above, and type III, class 2 gaskets are intended for use at temperatures down to -60°F (-51°C).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, and size required (see 1.2 and 3.3).
c. If first article samples are required (see 3.2).
d. Packaging requirements (see 5.1).

6.3 First article. When a first article is required, the contracting officer should provide specific guidance to offerors whether the first article(s) should be a preproduction sample, an initial production sample, a first production item, or a standard production item from the contractor's current inventory; the number of samples to be inspected as specified in 4.2, and (when applicable) the specific tests to be performed on each sample. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Acceptance criteria. The acceptance criteria below are recommended for use. The acceptance criteria as specified in the contract or purchase order shall be binding. Unless otherwise specified, the following acceptance criteria are in accordance with ASQ Z1.4.

6.4.1 For end item visual examination. An acceptable quality level (AOL), expressed in terms of defects per hundred units, of 2.5 is recommended.

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6.4.2 For end item dimensional examination. An AOL, expressed in terms of defects per hundred units, of 4.0 is recommended.

6.5 Subject term (key word) listing.

55-gallon drum
Gas can
Seal

6.6 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a material quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoD 4140.27-M, *Shelf-life Management Manual*. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27-M, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Services and Agency administrators for the DoD Shelf-Life Program. Apprropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.hq.dla.mil/.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army-GL
Navy - AS
Air Force - 99

Preparing activity:
DLA-IS

(Project 5330-2008-004)

Review activities:
Army – AR, AT, MR
Navy – CG, MC
Air Force - 71

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